METRIC

MIL-DTL-0032532(SH)
18 September 2015

DETAIL SPECIFICATION

LIFE RAFT, INFLATABLE, 25-PERSON MARK-7 MOD 2, AND 50-PERSON MARK-8 MOD 2

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND
1333 ISAAC HULL AVE SE
WASHINGTON NAVY YARD DC 20376-0001

IN REPLY REFER TO
4121
Ser 05S/348
SEP 1 8 2015

From: Commander, Naval Sea Systems Command (SEA 05, CHENG)

Subj: CHENG'S INTENT FOR IMPLEMENTATION OF MIL-DTL-0032532, LIFE RAFT, INFLATABLE, 25-PERSON MARK-7 MOD 2, AND 50-PERSON MARK-8 MOD 2

Ref: (a) MIL-DTL-0032532, Life Raft, Inflatable, 25-Person Mark-7 Mod 2, and 50-Person Mark-8 Mod 2

1. Purpose. The use of MIL-DTL-0032532 is approved as the design specification for contracting 25- and 50-person life rafts for U.S. Navy ships.

2. Discussion. Certified life rafts are a mandatory requirement for all U.S. Navy ships to get underway. These life rafts are intended for shipboard installation to be used in the event a ship has to be abandoned at sea. They are military unique including a military specific drop height, container, sea painter, outfitting, and re-certification periodicity. There is an urgent need for replacement life rafts due to an emerging block obsolescence challenge. More than 80% of 25-person (MK-7) life rafts will reach the end of their 20 year service life during the next five year contract cycle. Reference (a) design specification provides the latest NAVSEA requirements for 25- and 50-person life rafts. This specification will support the contracting activity (NAVSUP) efforts to procure new life rafts, and sustain the inventory at the level required to support the Fleet. A business case was developed that shows it is more cost effective to purchase new life rafts than to engineer and execute a service life extension program. NSN 4220-01-444-6260 (MK-7 life raft) and NSN 4220-01-475-3185 (MK-8 life raft) apply.

3. Point of Contact. For information pertaining to inflatable life rafts, please contact the Combatant Craft and Boats Technical Warrant Holder, Dean Schleicher, SEA 05D4, commercial: 757-462-3274, email: dean.schleicher@navy.mil.

L. B. FULLER
By direction

Affixed to: MIL-DTL-0032532

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This specification is approved for interim use by the Commander, Naval Sea Systems Command (NAVSEA). Other activities in the Department of Defense may use this interim document.

1. SCOPE

1.1 Scope. This specification covers the 25-person Mark-7 Mod 2 and 50-Person Mark-8 Mod 2 inflatable life rafts, including inflation system and survival gear, and packed within a rigid stowage container, ready for use as abandon ship survival equipment.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-191 - Textile Test Methods

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS27826 - Plug Assembly, (Sealing Clamp) Life Raft Repair

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

46 CFR 160.026 - Water, Emergency Drinking (in Hermetically Sealed Containers), for Merchant Vessels

46 CFR 160.043 - Jackknife (with Can Opener) for Merchant Vessels

46 CFR 160.054 - Kits, First-Aid, for Inflatable Liferafts

46 CFR 160.061 - Fishing Tackle Kits, Emergency, for Merchant Vessels

46 CFR 160.151 - Inflatable Liferafts (SOLAS)

46 CFR 164.018 - Retroreflective Material for Lifesaving Equipment

(Copies of these documents are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

583-5106577 - Container - Type II - Fiberglass Reinforced Plastic for 50 Person Inflatable Life Raft

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583-5111055 - Container, U.S.N. Life Raft, 25-Person

(Copies of these documents are available online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA TECHNICAL PUBLICATIONS

S9008-EA-INS-010 - Survival at Sea Instructions for the Operation and Use of the U.S. Navy 25-Person Life Raft (Mark 7) and 50-Person Life Raft (Mark 8)

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D413 - Standard Test Methods for Rubber Property-Adhesion to Flexible Substrate

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D751 - Standard Test Methods for Coated Fabrics

ASTM D2261 - Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)

(Copies of these documents are available online at www.astm.org.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA G-7.1 - Commodity Specification for Air

(Copies of this document are available online at www.cganet.com.)

DEFENSE ACQUISITION UNIVERSITY

Nomograph of Cumulative Binomial Distribution – Sampling Chart

(Copies of this document are available online at https://acc.dau.mil/docs/dtepi/pns/doc/nomogr/bdi.pdf.)

IEEE/ASTM INTERNATIONAL

IEEE/ASTM SI10 - American National Standard for Metric Practice

(IEEE and ASTM International published this standard jointly. Copies are available from www.ieee.org or www.astm.org.)

INTERNATIONAL MARITIME ORGANIZATION

IMO Resolution MSC.48(66) Chapter I, Regulation 1.2 - General Requirements for Life-Saving Appliances

IMO Resolution MSC.48(66) Chapter IV, Regulation 4.1 - General Requirements for Life Rafts

IMO Resolution MSC.48(66) Chapter IV, Regulation 4.2 - Inflatable Life Rafts

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IMO Resolution MSC.81(70) - Revised Recommendation on Testing of Life-Saving Appliances

SOLAS V/16 - Safety of Navigation – Maintenance of Equipment

(Copies of these documents are available online at www.imo.org.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

NBS Special Publication 440 - Color: Universal Language and Dictionary of Names

(Copies of this document are available online at https://archive.org/details/NISTresearchlibrary.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Regulatory compliance. The life raft shall be an existing product or non-developmental item (NDI) which currently holds U.S. Coast Guard approval and can meet the requirements of this specification with no more than minor modifications. Except as specified herein, the inflatable life raft shall meet the requirements of Chapter IV of MSC.48(66) and 46 CFR 160.151. The inflatable life raft shall be constructed in accordance with the following the International Convention for the Safety of Life at Sea (SOLAS) 74/83, hereinafter referred to as SOLAS, provisions:

a. IMO Resolution MSC.48(66) Chapter I, Regulation 1.2

b. IMO Resolution MSC.48(66) Chapter IV, Regulation 4.1

c. IMO Resolution MSC.48(66) Chapter IV, Regulation 4.2

3.4 Design and construction.

3.4.1 Shape. The inflated life raft may be circular, hexagonal, octagonal, elliptical, or similar outline in plan view.

3.4.2 Resistance to deterioration. All materials used in the construction of the life raft shall be of good quality and suitable for the purpose intended, and shall not be subject to deterioration from the effects of weathering aboard ship under the conditions of usual stowage, nor from contact with salt water or spray or petroleum products. The life raft shall be resistant to deterioration where exposed to sunlight.

3.4.3 Stability. The life raft shall have water-capturing stability appendages on its underside to resist capsizing from wind and wave forces. These appendages shall meet the following requirements:

a. The total volume of the appendages shall be not less than 2.4 cubic meters (m^3). Any opening designed to close as water is forced out of an appendage is not considered an opening for the purpose of this calculation.

b. The appendages shall be securely attached and evenly distributed around the periphery of the exterior bottom of the life raft. The appendages may be omitted at the locations of inflation cylinders.

c. The system of stability appendages shall consist of at least two separate parts so that damage to one part will permit at least half of the required total volume 1.2 m^3 to remain intact.

d. Openings in or between the stability appendages shall be provided to limit the formation of air pockets under the life raft.

e. The appendages shall be designed to deploy underwater when the life raft inflates. If weights are used for this purpose, they shall be of non-corrodible material or suitably protected against corrosion.

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3.4.4 Body.

3.4.4.1 Buoyancy chambers. Two main buoyancy chambers shall be superimposed, one atop the other, each inflated through a non-return inflation valve on each chamber. The buoyancy chambers shall be arranged so that, in the event of any one of the chambers being damaged or failing to inflate, the intact compartments shall be able to support, with positive freeboard over the life raft's entire periphery, 25 persons (Mark-7) or 50 persons (Mark-8), each having a mass of 75 kilograms and seated in their normal positions.

3.4.4.2 Storage pockets. Suitable pockets or equivalent shall be fitted for stowage of hand pumps, repair kits, instruction manuals, and other equipment needed immediately on launching.

3.4.4.3 Future equipment stowage pocket. A pocket to be used for future equipment shall be located on the side of the arch tube so as not to strike a seated occupant. The pocket shall be sized to hold a rectangular shape, 23 centimeters (9 inches) tall, 10 centimeters (4 inches) wide, and 5 centimeters (2 inches) thick, with a weight capacity of up to 1.13 kilograms (2.5 pounds) and the opening on the top end. The pocket shall have a flap that closes the opening and is secured closed with two pairs of 1.27-centimeter (½-inch) wide twill tape ties.

3.4.4.4 Reinforcing patches. Reinforcing patches shall be substantially fitted in way of the attachments for the righting line, painter, sea anchor line, etc.

3.4.4.5 Inflation cylinder holders. The holders for retaining the inflation cylinders shall be substantially fitted and sufficiently strong to retain the cylinders when the life raft is inflated or when the life raft is dropped into the water.

3.4.4.6 Exposed surfaces. All exposed surfaces shall be reasonably smooth and free from sharp protrusions or projections that might be injurious in boarding or occupancy.

3.4.5 Floor. The floor of the life raft shall be waterproof and shall be capable of being sufficiently insulated against cold by means of one or more compartments which the occupants can manually inflate, or which inflate automatically and can be deflated and re-inflated by the occupants. Insulation by other equally efficient means not dependent on inflation may also be used.

3.4.6 Canopy. The life raft design shall incorporate an inner and outer canopy to insulate the occupants from the weather and which shall automatically erect upon initial inflation of the life raft. Means shall be provided to prevent accumulation of water in the air gap. The canopy shall provide adequate headroom and provision shall be made for furling the ends to allow entry of occupants, free passage of air, and adequate room for paddling. Provision for catching rain water shall be incorporated onto the canopy. The canopy shall be provided with at least one viewing port for the Mark-7 and two viewing ports for the Mark-8. The interior of the canopy shall be of a color that does not cause discomfort to the occupants. Each entrance shall be clearly indicated and be provided with efficient adjustable closing arrangements which can be easily and quickly opened from inside and outside the life raft so as to permit ventilation, but exclude seawater, wind, and cold. The life raft shall have at least two diametrically opposite entrances.

3.4.7 Laps and seams. Each seam shall be at least as strong as the weakest of the materials joined by the seam. Each seam shall be covered with tape where necessary to prevent lifting of and damage to fabric edges. Vulcanization shall not be used. Chemically cured adhesives and radio frequency (RF) welded (polyurethane coated fabrics) methods are acceptable and shall meet the requirements of 3.10.15. There shall be no sewing that penetrates the fabric of an air retaining chamber.

3.4.8 Liners. A protective liner or baffling arrangement shall be provided inside each automatically inflatable compartment at the inflation gas inlet to protect the compartment fabric from the damaging effects of cold inflation gas.

3.4.9 Dissimilar materials. Where dissimilar materials are joined in the construction of the inflatable life raft, provisions shall be made to prevent loosening or tightening due to differences in thermal expansion, freezing, buckling, galvanic corrosion, or other incompatibilities.

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3.4.10 Retro-reflective material. The life raft shall be marked with Type I retro-reflective material approved under 46 CFR 164.018. Retro-reflective materials shall be fitted around the canopy of the life raft. The material shall be sufficiently wide and long to give a minimum area of 150 square centimeters (cm^2) and shall be spaced at suitable intervals (approximately 80 millimeters from center to center) at a suitable height above the waterline, doorways included, if suitable. Retro-reflective materials shall also be fitted to the underside of the floor, cross-shaped in the center. The dimension of the cross shall be half the diameter of the life raft and a similar cross shall be applied to the top of the canopy.

3.4.11 Color. The primary color of the exterior of the canopy shall be vivid reddish orange (color number 34 of National Institute of Standards and Technology NBS Special Publication 440), or a fluorescent color of a similar hue. The remainder of the life raft may be the natural color of the finished coated fabric, except that the underside of the floor shall be dark blue, dark grey, or black.

3.4.12 Container. The life raft containers shall be constructed in accordance with NAVSEA Drawing 583-5111055 for the Mark-7 life raft and NAVSEA Drawing 583-5106577 for the Mark-8 life raft. The container shall be held closed with ASTM B36 brass bands, 1.9 centimeters (¾ inch) wide, 0.05 centimeter (0.020 inch) thick, which are secured closed with two CRES securing clips on each band; two bands on the Mark-7 container or four bands on the Mark-8 container. The life raft shall be packed in its container in such a way as to ensure that the waterborne life raft inflates in an upright position on breaking free from its container.

3.4.13 Inflation. Inflation shall take place upon the pulling of the painter line, which may be accomplished manually both from the deck of a vessel or by a swimmer in the water. In addition, inflation connections shall be provided for use with the manually operated pump or bellows. Refer to 3.6 for the inflation system requirements.

3.4.14 Workmanship and materials. The life raft shall be constructed with proper workmanship and materials. A multiplicity of pin holes in any particular section is not considered repairable. Treating or swashing the inflatable members of the life raft with latex or any other sealing agents is not permissible. Repairs or rework shall be as specified (see 6.2). The ends of all ropes and webbing shall be fused, whipped, or otherwise secured to prevent fraying. Life raft coated fabric materials shall meet the performance requirements of 3.10.

3.4.15 Temperature ranges. The life raft shall not be damaged in stowage throughout an air temperature range of -30 to +65 °C. The life raft shall operate throughout a seawater temperature range of -1 to +30 °C.

3.5 Life raft fittings. The life raft shall be provided with the following fittings.

3.5.1 Boarding ramp. On the Mark-7 life raft, at least one entrance shall be fitted with a semi-rigid inflatable boarding ramp. The Mark-8 shall have at least two entrances fitted with boarding ramps. The boarding ramp shall have sufficient size and buoyancy to support one person of 100 kilograms mass in a sitting or kneeling position without holding onto any other part of the life raft. The boarding ramp shall be arranged so as to prevent deflation of the life raft if the ramp is damaged.

3.5.2 Boarding ladder. Entrances not provided with a boarding ramp shall have a boarding ladder, the lowest step of which shall be situated not less than 400 millimeters below the life raft's light waterline. The steps of the boarding ladder shall be of rigid or semi-rigid construction and secured against rotation to provide a suitable foothold. There shall be means inside the life raft to assist persons to pull themselves into the life raft from the ladder.

3.5.3 Exterior canopy lamp. The Mark-7 life raft shall be equipped with one exterior canopy lamp; the Mark-8 with two. The lamp shall be visible from the surface of the sea from any direction on a dark night with a clear atmosphere at a distance of at least 3.2 kilometers for a period of not less than 12 hours. The light shall flash at a rate of not less than 50 flashes per minute for the first 2 hours of operation of the 12-hour operating period. The lamp shall use light-emitting diodes (LEDs) in lieu of incandescent bulbs. The lamp shall be powered by replaceable lithium cell battery(ies) and shall light automatically when the life raft inflates. The lamp shall be fitted with a manual control to allow conservation of power during daylight hours. The battery(ies) shall be sufficiently protected from the dampness or humidity in the stowed life raft.

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3.5.4 Interior canopy lamp. The Mark-7 shall have one interior canopy lamp; the Mark-8 shall have two. A manually controlled lamp shall be fitted inside the life raft. The lamp shall use LEDs in lieu of incandescent bulbs and be powered by a replaceable lithium battery(ies) that is sufficiently protected from the dampness or humidity in the stowed life raft. The lamp shall be capable of continuous operation for a period of at least 12 hours. It shall light automatically when the life raft inflates and be of sufficient intensity to enable reading of survival and equipment instructions.

3.5.5 Lifelines. One lifeline shall be fitted around the outside periphery and one around the inside of the life raft. The two lifelines shall be made of 14-millimeter minimum width nylon tubular webbing, or 10-millimeter minimum diameter rope, or equivalent. Each lifeline attachment patch shall have a minimum breaking strength of 1.5-kilonewton (kN) pull exerted in a direction perpendicular to the base of the patch. Each bight of the exterior lifeline shall be long enough to allow the lifeline to reach to the waterline of the life raft when it is afloat.

3.5.6 Sea painter system. A sea painter with bung assembly shall be provided for inflating and holding the life raft during boarding. It shall be a sea painter with bung assembly, part number HRC-950, from Hampton Rubber Co., or equal. The sea painter, with an approximate breaking strength of 181 kilograms, shall act as the weak link. The sea painter and rubber plug shall be installed as shown on figure 1.

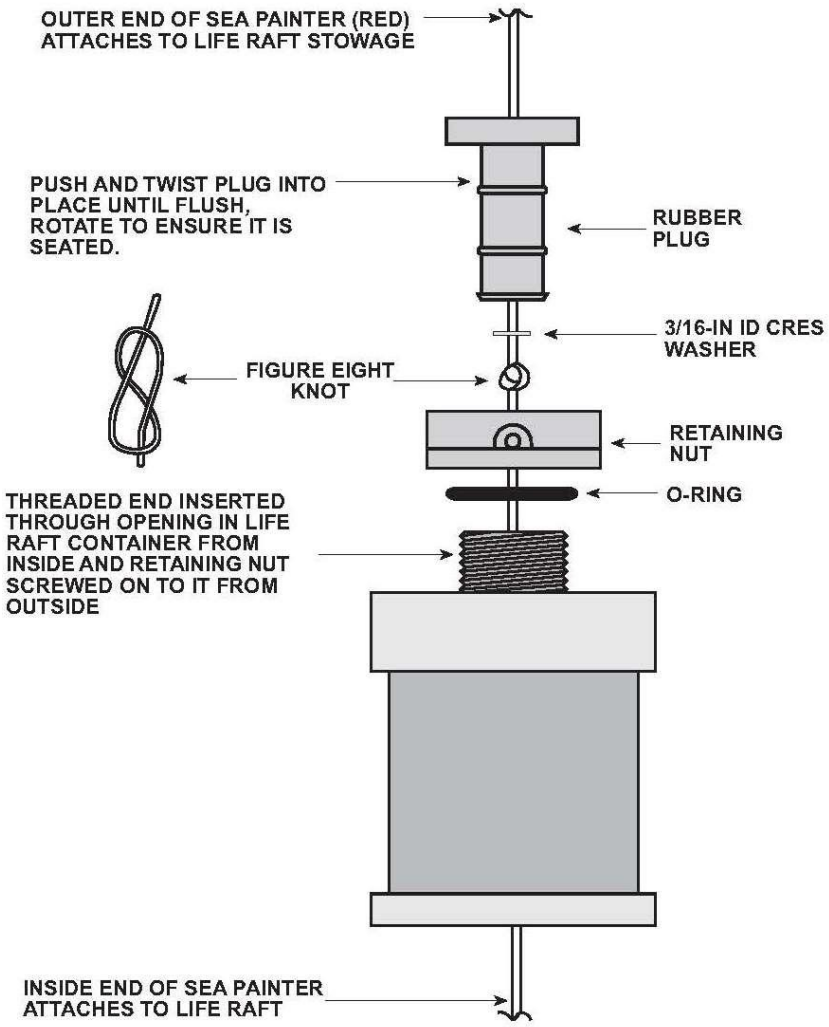


FIGURE 1. Sea painter/bung assembly arrangement.

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3.5.7 Righting gear. Suitable handholds or straps on the underside of the floor shall be included to enable a righting moment to be exerted. Such straps shall be webbing of adequate strength to allow righting the inverted life raft. The righting strap shall run the full width of the life raft and shall be secured to the buoyancy chamber on both sides of the life raft. The stability of the life raft when in the inverted position shall be such that it can be righted in a sea way and in calm water by one person.

3.5.8 Towing/mooring connections. Four towing/mooring connections shall be provided; one each at opposite ends of the life raft on the lower inflatable hull tube section, and two connections spaced equidistant on the lower hull tube between those two fittings. The mooring/towing connections shall be marked to indicate their function. The life raft and its fittings shall be constructed so as to enable it to be towed at a speed of 3 knots in calm water when loaded with its full complement of persons and equipment and with one of its sea anchors deployed.

3.6 Inflation system. The life raft shall be equipped with an automatic inflation system, which consists of a commercial off-the-shelf (COTS) standard Department of Transportation (DOT)-approved carbon fiber over aluminum composite cylinder, a shop repairable inflation valve, and a multi-use inflation hose with end fittings compatible with the inflation valve and the raft inlet check valve. The inflation gas shall be Grade D breathable air in accordance with CGA G-7.1, with a maximum pressure of 31 megapascals (MPa) at 21 °C. The inflation valve shall be serviceable by life raft shop personnel with appropriate rebuild kit available.

3.6.1 Inflation cylinders. Inflation cylinders shall be carbon fiber wrapped aluminum composite air cylinders with a minimum service life of 15 years. The Mark-7 life raft inflation cylinders shall have a female valve port sized to accept an inflation valve with 0.875-14 UNF (male) threads and an O-ring mating surface such that a minimum of six threads are engaged and there is a maximum working pressure of 4,500 pounds per square inch (psi). The Mark-7 cylinder shall have a volume of 9,013 centimeters cubed (cm^3) (550 cubic inches [in^3]), a diameter of 17.78±0.635 centimeters (7±0.25 inches), and an overall length of 54.61±1.27 centimeters (21.5±0.5 inches). The Mark-8 composite inflation cylinders shall be sized as necessary to meet inflation requirements of the life raft in all stated temperature extremes. Each compressed gas inflation cylinder within the life raft shall be installed as follows:

a. The slings and reinforcements shall be of sufficient strength to retain the inflation cylinders in place when the life raft is dropped into the water from a height of 23 meters during deployment.

b. The painter and inflation cylinders of the life raft are linked to start inflation when the painter is pulled by one person exerting a force not exceeding 150 Newtons (N).

3.6.2 Inflation valve. The inflation valve shall be equipped with a safety relief feature which will release the cylinder pressure when the cylinder reaches a temperature of 75+2/-0 °C.

3.6.3 Pressure relief valve poppet retainer. Poppet retainers for rendering pressure relief valves inoperable shall be provided for every pressure relief valve installed in the life raft, unless the pressure relief valves are of a type that can be rendered inoperable. If provided, pressure relief valve poppet retainers shall either float or be secured to the life raft by a lanyard.

3.6.4 Valves. The pressure relief valves, inlet check valves, and topping off/dumping valves shall fit in a hull tube-mounted flange system with a 1.312016 UN-2B thread opening. The flanges shall be mounted in the hull tubes, arch tube, and any other inflatable section requiring these types of valves. The flanges shall be adhered to the inflatable sections using RF welding for polyurethane coated fabrics or appropriate adhesive for the materials being joined. The relief valves shall relieve the pressure at 190±10 percent of the designated working pressure and will reseat at 120±10 percent of the designed working pressure. Relief valves shall be as small as practical and shall be mounted on a flat molded 127-millimeter diameter universal flange that can be easily attached to the life raft flotation tubes. All metal parts that can cause damage to the life raft fabric shall be covered with molded neoprene.

3.7 Life raft equipment. The following equipment shall be provided:

3.7.1 Buoyant rescue quoit and heaving line. A buoyant rescue quoit and heaving line not less than 30 meters in length and having a breaking strength of not less than 1.1 kN shall be provided and shall be attached to the life raft near the entrance furthest from the painter attachment.

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3.7.2 Knife. Two knives shall be provided. One knife of the non-folding type having a buoyant handle and lanyard attached shall be stowed in a pocket on the exterior of the canopy near the point at which the painter is attached, and one of the folding type in accordance with 46 CFR 160.043, which shall be stowed in the survival equipment bag or bags.

3.7.3 Bailers. One or two bailers having a total aggregate volume of at least 2 liters shall be provided. Each bailer shall be equipped with a lanyard which is at least 1 meter in length.

3.7.4 Cellulose sponges. Two cellulose sponges, each having a volume of at least 750 cm^3 when saturated with water, shall be provided.

3.7.5 Sea anchor. Two sea anchors, each with a shock-resistant hawser and tripping line, shall be provided. One sea anchor, being the primary sea anchor, shall be permanently attached to the life raft in such a way that, when the life raft inflates or is waterborne, it will cause the life raft to lie oriented to the wind in the most stable manner, and the other, being a spare sea anchor, shall be packed in the survival equipment. The strength of each sea anchor and its hawser and tripping line shall be adequate for all sea conditions. The sea anchors shall be fitted with a swivel at each end of the line and shall be of a type which is unlikely to turn inside-out between its shroud lines. Sea anchors without the swivel may be used if, during the towing test, their design is demonstrated to be of a type that does not rotate when streamed. The sea anchors are not required to have the tripping lines if, during the towing test, their design is demonstrated to be of a type that can be hauled in by one person.

3.7.6 Buoyant paddles. Two buoyant paddles of at least 1.2 meters long shall be provided. The paddles shall be of the same size and type as used to pass the maneuverability test of Part 1, 5.10 of IMO Resolution MSC.81(70).

3.7.7 Pump or bellows. A manually operated pump or bellows, with a connecting fitting compatible with the topping-off valve and capable of inflating any inflatable chamber to its working pressure, shall be provided. The Mark-7 life raft shall have at least one pump or bellows and the Mark-8 shall have at least two pumps or bellows.

3.7.8 First-aid kit. An approved first-aid kit(s) in accordance with 46 CFR 160.054 shall be provided. The Mark-7 shall have one first-aid kit and the Mark-8 shall have two.

3.7.9 Abandon ship signal kit. An abandon ship signal kit (NSN 1370-01-366-0344) shall be packed with the survival equipment. The kit shall contain six Mark-124 day/night signals and two Mark-79 aerial signal kits. The abandon ship signal kit shall be Government Furnished Material (GFM), as specified (see 6.2 and 6.3).

3.7.10 Electric flashlight. One electric LED flashlight shall be provided. It shall be capable of a minimum of 60 lumen output after 4 hours of use, be bright orange or yellow in color, and shall use no more than 4 AA 1.5-volt DC lithium batteries. The maximum brightness shall not exceed 120 lumens. It shall be waterproof to a depth of 2 meters. It shall have an on/off switch that allows for momentary signaling, such as Morse code signaling. It shall have a lanyard with a circumferential length of 1 meter (±5 centimeters) to hang about the neck.

3.7.11 Mirror, signaling. One daylight signaling mirror with instructions on its use for signaling to ships and aircraft shall be provided.

3.7.12 Whistle. One ball type whistle or multi-tone whistle, made of corrosion-resistant materials, shall be provided.

3.7.13 Lifesaving signal waterproof card. One copy of the lifesaving signals referred to in SOLAS V/16 shall be provided on a waterproof card or in a waterproof container.

3.7.14 Fishing tackle. One emergency fishing tackle kit shall be provided in accordance with 46 CFR 160.061.

3.7.15 Food rations. Each Mark-7 life raft shall be equipped with 25 food rations, and each Mark-8 with 50 food rations. Each food ration shall total not less than 10,000 kilojoules. These rations shall be kept in airtight packaging and shall be stowed in a watertight container. Rations shall be palatable, have a minimum shelf-life of 5 years, and be packed in a manner whereby it can be readily divided into daily portions and easily opened. Each ration shall have its contents packed into a minimum of 6 individually wrapped portions. Each ration shall be labeled “Ration for one person for three days” and shall have a United States Coast Guard (USCG) approval number.

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3.7.16 Drinking water. Each Mark-7 life raft shall be packed with 25 bottles of drinking water, and each Mark-8 with 50 bottles of drinking water. Each 500 ml plastic watertight bottle of potable water shall meet the requirements of 46 CFR 160.026. The bottle shall be graduated in 100-milliliter increments.

3.7.17 Manual reverse osmosis desalinator (MROD). One MROD (NSN 4610-01-319-5337) shall be packed with the survival equipment. The MROD shall be GFM, as specified (see 6.2 and 6.3).

3.7.18 Anti-seasickness tablets. Six doses of anti-seasickness medicine shall be provided for each person (Mark-7: 25 persons, Mark-8: 50 persons). The anti-seasickness medication required shall include instructions for use and shall be marked with an expiration date. The medication shall be either of the following:

a. A combination of 25 milligrams of promethazine hydrochloride and 25 milligrams of ephedrine sulfate, comprising each single dose, to be taken at 6-hour intervals; or

b. A transdermal patch containing scopolamine suitable for at least 2 days' use, with each patch counted as six doses.

3.7.19 Survival instructions. Each life raft shall be equipped with one hardcopy of S9008-EA-INS-010 (NSN 0910-LP-102-2272), which are the instructions on how to survive in a life raft. The instructions shall:

a. Be waterproof.

b. Be in English.

c. Be printed from a PDF format file provided as GFM, as specified (see 6.2 and 6.3).

d. Be suspended in a clear film envelope from one of the canopy arch tubes.

3.7.20 Immediate action instructions. Each life raft shall be equipped with one hardcopy of immediate action instructions. The instructions for immediate action shall:

a. Be waterproof.

b. Be in English.

c. Explain both the noise accompanying the operation of any provided pressure relief valves and the need to render them inoperable after they complete venting.

d. Be suspended from the inside canopy, so they are immediately visible by survivors upon entering the life raft. They may be contained in the same envelope with the instructions on how to survive, if the instructions for immediate action are visible from both faces of the envelope.

3.7.21 Thermal protective aid. Thermal protection shall be provided for the occupants of the life raft by aluminized Mylar thermal protective blankets. The Mark-7 shall be equipped with 12 each and the Mark-8 with 25 each thermal protective blankets. The thermal protective blanket shall:

a. Be made of waterproof material having a thermal conductivity of not more than 0.25 watt per square meter degree Kelvin.

b. Have a physical size of 2.15 by 1.4 meters, with a tolerance in both length and width of ±0.15 meter.

c. The coated material shall have a minimum tensile strength of 82.7 Newton/square millimeters (N/mm^2) and a thickness of 0.012 millimeter, -0.0 millimeter.

d. Be reflective, coated on both sides of the blanket.

e. Be packed in a waterproof package with desiccant pack.

3.7.22 Repair outfit. Repair kits shall be provided and installed in a pocket or survival equipment bag inside the life raft. The Mark-7 shall have two repair kits and the Mark-8 shall have four repair kits. Each kit shall:

a. Be sealed in a waterproof package with a desiccant pack to prevent moisture.

b. Contain one each of the following size sealing clamps in accordance with MS27826:

(1) 76-millimeter (3-inch) (MS27826-1)

(2) 127-millimeter (5-inch) (MS27826-2)

(3) 190-millimeter (7½-inch) (MS27826-3)

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3.7.23 Survival equipment bag or bags. The life raft shall be provided with a survival equipment stowage bag or bags of a waterproof material having a re-sealable, watertight opening; capable of being tied to the inside of the life raft with short lengths of nylon line, webbing, or equivalent; and sized to stow the following equipment:

a. Knife, Folding (see 3.7.2)
b. Bailer(s) (see 3.7.3)
c. Sponges (see 3.7.4)
d. First-Aid Kit (see 3.7.8)
e. Abandon Ship Signal Kit (see 3.7.9)
f. Electric Flashlight and Batteries (see 3.7.10)
g. Mirror, Signaling (see 3.7.11)
h. Whistle (see 3.7.12)
i. Fishing Tackle (see 3.7.14)
j. Food Rations (see 3.7.15)
k. Drinking Water, Bottled (see 3.7.16)
l. Manual Reverse Osmosis Desalinator (MROD) (see 3.7.17)
m. Anti-Seasickness Tablets (see 3.7.18)
n. Thermal Protective Aids (see 3.7.21)

3.8 Size and mass. The mass of the Mark-7 life raft, including its container and equipment, shall not exceed 185 kilograms (408 pounds.). The mass of the Mark-8 life raft, including its container and equipment, shall not exceed 340 kilograms (750 pounds).

3.9 Marking.

3.9.1 Life raft nameplate. The life raft shall have a permanently attached substantial nameplate of compatible material on which is embossed or imprinted with the following:

a. Life Raft, Inflatable
b. Mark-7 Mod 2, 25 Persons or Mark-8 Mod 2, 50 Persons (as applicable)
c. Stock No.
d. Manufacturer Name
e. Date of Manufacture (month, year)
f. Contract No. _____
g. Serial No. _____

3.9.2 Life raft container data plate. The life raft container shall have a non-ferrous metal data plate affixed to the top half of the life raft container with compatible non-corrosive pop rivets. The data plate shall have the following information with a minimum text height of 3 millimeters:

a. US NAVY LIFE RAFT
b. MK-7 MOD 2, 25 PERSONS or MK-8 MOD 2 50 PERSONS (as applicable)
c. MFR (plus name of manufacturer)
d. S/N (plus serial number)
e. CERT DATE (plus date of certification)
f. EXP DATE (plus date of certification expiration)
g. FLARE LOT NUMBER (plus lot number from GFM signal kit)
h. FAC (plus certification facility)

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3.9.3 Serial numbers. An 8-character, alpha-numeric serial number shall be assigned to each life raft by the Contractor. The life raft shall have a unique, systematic serial number regardless of contracts or purchase orders. The serial number shall consist of three characters representing the manufacturer, a space followed by a 5-digit sequential number starting with 00001 (e.g., EXP 00001).

3.9.4 Marking. Marking shall be in English and clearly and legibly applied in a color contrasting to its background, using materials which are permanent for the life of the life raft. Markings shall be located appropriate to the action indicated. Markings shall include:

a. Instructions for manual inflating.

b. Directions for righting if the life raft inflates in an inverted position.

c. Directions for boarding.

d. Position and use of items stowed outside of the equipment container.

e. Contents of survival equipment container(s).

f. “25 PERSONS” or “50 PERSONS”, as applicable, shall be marked over each entrance in characters not less than 100 millimeters in height.

3.9.5 Code symbols. International Morse Code for the English alphabet A through Z and for Arabic numerals 0 through 9 shall be printed on a 200- by 300-millimeter area or on a piece of canopy fabric and cemented on the inside of the raft canopy adjacent to one entrance. Each code symbol shall be identified by its appropriate letter or numeral counterpart next to the symbol.

3.10 Coated fabric. If a coated fabric different from the fabric used on the USCG-approved commercial 25- or 50-person life rafts is to be used on the proposed life raft design, the coated fabric used in the construction of inflatable buoyancy chambers shall meet the requirements of this section. Polyvinyl chloride (PVC)-coated fabric or PVC alloy-coated fabric shall be prohibited, regardless of USCG or SOLAS certification. The base fabric shall be inherently rot proof and shall not be unduly affected by seawater, oil, or fungal attack. Water soluble materials shall not be used in the coating compound, which shall be applied in a manner such that the finished product consistently meets the performance requirements of this section.

3.10.1 Breaking strength. The breaking strength of the finished coated fabric shall be a minimum of 53 N/mm in warp and fill when tested in accordance with 4.6.1.

3.10.2 Tearing strength. The tearing strength of the finished coated fabric shall be a minimum of 72 Newtons in warp and fill when tested in accordance with 4.6.2.

3.10.3 Coating adhesion. The coating adhesion shall be determined by using a suitable air-cure adhesive to bond one face of the coated fabric to the other face. The minimum adhesion shall be 2.8 N/mm with the coating compound separating off the base fabric when tested in accordance with 4.6.3.

3.10.4 Weld strength of thermoplastic coated fabrics. The coated fabric shall have a minimum peel load of 7 N/mm for the initial break in accordance with 4.6.4.

3.10.5 Resistance to blocking (heat). The coated fabric shall separate in 5 seconds or less when tested in accordance with the procedures of 4.6.5

3.10.6 Porosity test. The coated fabric shall show no signs of bubble formation indicating leakage when subjected to a maximum pressure of 70 kilo Pascal gauge (kPag) held for 5.0±0.1 minute when tested in accordance with 4.6.6.

3.10.7 Coating adhesion after exposure to water. A minimum adhesion result shall be 1.75 N/mm of width when tested in accordance with 4.6.7.

3.10.8 Puncture. The coated fabric shall have a puncture resistance of 360 Newtons when tested in accordance with 4.6.8

3.10.9 Fungus. The coated fabric shall maintain a minimum of 80 percent of original breaking strength and a coating adhesion of 1.75 N/mm when subjected to fungus in accordance with 4.6.9.

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3.10.10 Resistance to oil. When sample is tested in accordance with 4.6.10, a single pass of a finger across the sample shall not produce smearing.

3.11 Performance requirements.

3.11.1 Inflation. The Mark-7 and Mark-8 life rafts shall be automatically inflated upon pulling the sea painter. Inflation shall be completed within a period of 30 seconds at an ambient temperature of 18 to 20 °C when tested in accordance with 4.7.3. In addition, inflation time shall be within a period of 30 seconds for the Mark-7 and 60 seconds for the Mark-8, at an ambient temperature of -30 °C when tested in accordance with 4.7.4. After inflation, the life raft shall maintain its form when loaded with its full complement of persons and equipment.

3.11.2 Over-pressure. The inflatable compartment shall be capable of withstanding a pressure equal to at least 3 times the working pressure and shall be prevented from reaching a pressure exceeding twice the working pressure by means of relief valves when tested in accordance with 4.7.5.

3.11.3 Environmental exposure. The life raft shall be constructed so as to be capable of withstanding exposure for 30 days afloat in all sea conditions when tested in accordance with 4.7.6.

3.11.4 Operation when dropped. The life raft, in its container, shall be constructed such that when it is dropped into the water from a height of 23 meters, it shall remain intact and float for a period of 30 minutes. The life raft, when inflated by means of the painter line, shall inflate upright and be boardable when tested in accordance with 4.7.11. The life raft and its equipment shall operate satisfactorily.

3.11.5 Withstanding of jumps. The floating life raft shall be capable of withstanding 25 jumps for the Mark-7 or 50 jumps for the Mark-8 onto it from a height of at least 4.5 meters above its floor. This shall consist of 12 jumps (Mark-7) or 25 jumps (Mark-8) without the canopy erected and 13 jumps (Mark-7) or 25 jumps (Mark-8) with the canopy erected when tested in accordance with 4.7.2. Persons jumping shall weigh no less than 75 kilograms.

3.11.6 Towing. The stability of the life raft, when loaded with its full complement of persons and equipment, shall be such that it can be towed at speeds of up to 3 knots in calm water for a distance of 1 kilometer when tested in accordance with 4.7.10.

3.11.7 Service life. The life raft (exclusive of ancillary equipment) and its container shall have an expected service life of at least 20 years when recertified every 5 years and maintained according to manufacturer's instructions.

3.11.8 Air retention. Air pressure drop shall not exceed 2.5 percent after compensation for temperature and barometric pressure change when tested in accordance with 4.7.1.

3.11.9 Loading and seating. The loading and seating shall be determined using the procedure in 4.7.7 to verify the Mark-7 life raft can accommodate 25 people and the Mark-8 life raft can accommodate 50 people.

3.11.10 Canopy closure. There shall be no more than 20 liters of accumulated water when tested in accordance with 4.7.8.

3.11.11 Lift out force. The peak force measured when lifting the life raft out of the water shall be at least 3,315 Newtons when tested in accordance with 4.7.9.

3.12 Servicing.

3.12.1 Frequency. The life raft shall be designed to be serviced once every 5 years at approved Navy servicing facilities.

3.12.2 Service manual. A service manual covering the opening, inspection, testing, repair, and repacking of the approved life raft with parts list of all replaceable/repairable components shall be provided, as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

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b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one life raft, complete with equipment and stowage container, when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.7.1 through 4.7.11, as indicated in table I.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4, 4.5, and the tests of 4.7.1 through 4.7.11, as indicated in tables I through IV. Reliability based on sampling shall be 80 percent with a minimum confidence level of 0.5 in accordance with the Cumulative Binomial Distribution Nomograph.

TABLE I. First article tests and conformance inspections.

Characteristic	Requirement Paragraph	Test Paragraph	First Article Test	Conformance Inspection	Verification Method
Shape	3.4.1	4.4	X		Visual Inspection
Stability	3.4.3	4.4	X		Visual Inspection & Certificate of Compliance
Buoyancy Chambers	3.4.4.1	4.4	X		Visual Inspection
Storage Pockets	3.4.4.2	4.4	X		Visual Inspection
Future Equipment Pocket	3.4.4.3	4.4	X		Visual Inspection
Reinforcing Patches	3.4.4.4	4.4	X	X	Visual Inspection
Inflation Cylinder Holders	3.4.4.5	4.4	X		Visual Inspection
Exposed Surfaces	3.4.4.6	4.4	X	X	Visual Inspection
Floor	3.4.5	4.4	X	X	Visual Inspection
Canopy	3.4.6	4.4	X	X	Visual Inspection
Laps and Seams	3.4.7	4.4	X	X	Test Documentation
Liners	3.4.8	4.4	X		Design Documentation
Retro-Reflective Material	3.4.10	4.4	X	X	Visual Inspection
Color	3.4.11	4.4	X	X	Visual Inspection
Container	3.4.12	4.4	X	X	Visual Inspection
Inflation	3.4.13	4.4	X	X	Demonstration
Boarding Ramps	3.5.1	4.4	X	X	Visual Inspection
Boarding Ladder	3.5.2	4.4	X	X	Visual Inspection
Exterior Canopy Lamp	3.5.3	4.7.3	X		Demonstration & Test Documentation
Interior Canopy Lamp	3.5.4	4.7.3	X		Demonstration & Test Documentation
Life Lines	3.4.5	4.4	X	X	Visual Inspection & Test Documentation
Sea Painter System	3.5.6	4.4	X	X	Visual Inspection
Righting Gear	3.5.7	4.4	X	X	Visual Inspection
Towing/Mooring Connections	3.5.8	4.4	X	X	Visual Inspection & Test Documentation

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TABLE I. First article tests and conformance inspections – continued.

Characteristic	Requirement Paragraph	Test Paragraph	First Article Test	Conformance Inspection	Verification Method
Inflation Cylinders	3.6.1	4.4	X	X	Visual Inspection & Test Documentation
Inflation Cylinders	3.6.1.a	4.4	X		Test Documentation
Inflation Cylinders	3.6.1.b	4.7.3	X	X	Demonstration
Inflation Valve	3.6.2	4.4	X		Test documentation
Pressure Relief Valve Poppet Retainer	3.6.3	4.7.5	X		Demonstration
Valves	3.6.4	4.4	X	X	Visual Inspection & test documentation
Buoyant Rescue Quoit and Heaving Line	3.7.1	4.4	X	X	Visual Inspection
Knife	3.7.2	4.4	X	X	Visual Inspection
Bailers	3.7.3	4.4	X	X	Visual Inspection
Cellulose Spongers	3.7.4	4.4	X	X	Visual Inspection
Sea Anchor	3.7.5	4.4	X	X	Visual Inspection
Buoyant Paddles	3.7.6	4.4	X	X	Visual Inspection
Pump or Bellows	3.7.7	4.4	X	X	Visual Inspection
First Aid Kit	3.7.8	4.4	X	X	Visual Inspection
Abandon Ship Signal Kit	3.7.9	4.4	X	X	Visual Inspection
Electric Flashlight	3.7.10	4.4	X	X	Visual Inspection & Test Documentation
Mirror, Signaling	3.7.11	4.4	X	X	Visual Inspection
Whistle	3.7.12	4.4	X	X	Visual Inspection
Life-Saving Signal Table	3.7.13	4.4	X	X	Visual Inspection
Fishing Tackle	3.7.14	4.4	X	X	Visual Inspection
Food Rations	3.7.15	4.4	X	X	Visual Inspection & USCG Certification
Drinking Water	3.7.16	4.4	X	X	Visual Inspection & USCG certification
Manual Reverse Osmosis Desalinator (MROD)	3.7.17	4.4	X	X	Visual Inspection
Anti-Sea Sickness Tablets	3.7.18	4.4	X	X	Visual Inspection
Survival Instructions	3.7.19	4.4	X	X	Visual Inspection
Immediate Action Instructions	3.7.20	4.4	X	X	Visual Inspection
Thermal Protective Aid	3.7.21	4.4	X	X	Visual Inspection & Test documentation

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TABLE I. First article tests and conformance inspections – continued.

Characteristic	Requirement Paragraph	Test Paragraph	First Article Test	Conformance Inspection	Verification Method
Repair Outfit	3.7.22	4.4	X	X	Visual Inspection
Survival Equipment Bags	3.7.23	4.4	X	X	Visual Inspection
Size and Mass	3.8	4.4	X		Test documentation
Life Raft Nameplate	3.9.1	4.4	X	X	Visual Inspection
Life Raft Container Data Plate	3.9.2	4.4	X	X	Visual Inspection
Serial Numbers	3.9.3	4.4	X	X	Visual Inspection
Marking	3.9.4	4.4	X	X	Visual Inspection
Code Symbols	3.9.5	4.4	X	X	Visual Inspection
Coated Fabric (if applicable)	3.10.1 – 3.10.10	4.5 and 4.6 (if applicable)	X		Test Documentation
Inflation	3.11.1	4.7.3	X		Demonstration & Test Documentation
Over-Pressure Capability	3.11.2	4.7.5	X		Test Documentation
Environmental Exposure	3.11.3	4.7.2	X		Test Documentation
Operation When Dropped	3.11.4	4.7.11	X		Test Documentation
Withstanding of Jumps	3.11.5	4.7.2	X		Test Documentation
Towing	3.11.6	4.7.10	X		Test Documentation
Service Manual	3.13.2	4.4	X		Visual Inspection
Typical Life Raft Defects	Table II	4.4		X	Visual Inspection
Air Retention	3.11.8	4.7.1	X	X	Test Documentation
Loading and Seating	3.11.9	4.7.7	X		Test Documentation
Canopy Closure	3.11.10	4.7.8	X		Test Documentation
Lift-Out Force	3.11.11	4.7.9	X		Test Documentation

4.4 Examination. Each life raft shall be examined for compliance with the requirements specified in 3.2 through 3.10 using table II. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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TABLE II. Types of typical life raft defects.

TYPICAL LIFE RAFT DEFECTS
1. Hoses damaged. End connections not torqued as specified in Service Manual. End connections cross-threaded.
2. Evidence of sewing in the seams of, or through the fabric of air chambers as specified in 3.4.7.
3. Valve pull-cables damaged or not connected as specified in Service Manual.
4. Evidence of treating or swashing inflatable member with latex or other sealing agent as specified in 3.4.14.
5. Topping-off valves not installed correctly as specified in 3.6.4.
6. Inflation cylinders not fully charged to the weight and pressure specified in Service Manual.
7. Banding straps not installed as specified in 3.4.12.
8. Painter rope missing, not attached, not as specified in Service Manual.
9. Pull force on painter line to activate the inflation system exceeds 150 Newtons as specified in 3.6.1.b.
10. Hand pump or bellows and adapters not operable as specified in 3.7.7.
11. Boarding ladder damaged, not as specified in 3.5.2.
12. Canopy damaged, not as specified in 3.4.6.
13. Sea anchors missing, not installed or stowed as specified in 3.7.5.
14. Cylinder holders not installed as specified in 3.4.4.5.
15. Weight of the completed, packed life raft, ready for stowage exceeds maximum weight specified in 3.8.
16. Materials used other than those specified. Survival gear missing or not as specified in 3.7.
17. Equipment incomplete, components not as specified, not stowed as specified in 3.7.
18. Length and width of the finished life raft not within the tolerances specified in Service Manual.
19. Hull, gunwale, cross tube, diameters not within the tolerances specified in Service Manual.
20. Unauthorized repairs or rework (see 3.4.14.).
21. Paddles damaged or missing, not as specified in 3.7.6.
22. Lights not installed as specified in 3.5.3 and 3.5.4, batteries missing.
23. Canopy closure fasteners not installed, not as specified in 3.4.6.
24. Righting gear missing, not installed as specified, not as specified in 3.5.7.
25. Floatable knives missing, not as specified in 3.7.2.
26. Inner canopy vent holes not as specified in 3.4.6.
27. Identification marking not legible, not as specified in 3.9.1.
28. Instruction cards missing, not as specified in 3.7.19.
29. Caution tag or instruction lettering missing, not as specified in 3.7.20.
30. Rope or line ends not fused or whipped as specified in 3.4.14.
31. Rain catcher as specified in 3.4.6 is missing.
32. Inflatable/Insulated floor not as specified in 3.4.5.
33. Sealing clamps missing or not as specified in 3.7.22.
34. Rescue heaving quoit and line not as specified, not installed as specified in 3.7.1.
35. Lifeline not as specified, not installed as specified in 3.5.5.

4.5 Material inspection.

a. Examination of the coated and cured fabric. Defects found during the examination shall be classified in accordance with item f below and table III.

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b. Meter-by-meter examination. The required meterage of each roll shall be inspected and visual defects classified as listed in item f below. The material shall be examined on one side; however, the side examined shall be alternated from roll to roll. The material shall be through-lighted and examined for pin holes. The defects shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the fabric, in which case only the more serious defects shall be counted. A continuous defect shall be counted as one defect for each warp-wise meter or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear meter. The number of pieces from which the sample is to be selected shall be in accordance with table IV. An approximate equal number of meters shall be examined in each piece of the sample.

c. Meter-by-meter fabric examination defects. A list of defects that may be present are listed in table III.

d. Overall examination. Each defect listed in table III shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size shall be in accordance with table IV. The lot shall be unacceptable if one or more of the defects listed are found:

e. Overall examination defects.

(1) Wrong color.

(2) Not clean throughout.

(3) Adhering to successive layer on roll (resists unrolling), tacky, cracked, or brittle.

(4) Objectionable odor.

(5) Variation in weight. The variation in mass shall be not greater than ±50 grams per meter squared (g/m^2) of the approved coated fabric.

TABLE III. Meter-by-meter fabric examination defects.

EXAMINATION DEFECTS	
Abrasion mark	Resulting in a weak place or evidence of coating removed (clearly visible at normal inspection distance (about 1 meter))
Blister lump or tunnel	Any soft (hollow), clearly noticeable
Crease or wrinkly embedded	Or resulting in doubling, or adhesion of surfaces against each other, or incomplete coating
Cut (including cuts through coating only), hole, tear, or pinhole	Any light areas or windows (poor distribution of pigments)
Uneven coating	Thin areas where coating compound is missing or insufficient; or heavily coated areas, noticeably stiffer than adjoining, unaffected fabric
Selvedge	Scalloped, deepest indentation 15 millimeters or more: tight, resulting in pucker or waviness in adjoining fabric
Uneven tension	Puckers, waviness, ridges that will not flatten under manual pressure
Spot, stain, or streak	Other than coating or pigment and more than 15 millimeters in its longest dimension
Width	Less than specified

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TABLE IV. Sampling for examination.

LOT SIZE (METERS)	SAMPLE SIZE (IN PIECES)
Up to 1,300	3
1,301 to 3,200	5
3,201 to 8,000	8
8,001 to 22,000	10
22,001 and up	15
NOTE: 1. If lot contains fewer than 3 pieces, each piece of that lot shall be examined.	

4.6 Coated fabric. The following tests shall be conducted when a coated fabric different from the fabric used on the USCG-approved commercial 25- or 50-person life rafts is used, as specified in 3.10.

4.6.1 Breaking strength. The finished coated fabric shall be tested in accordance with Method 5102 of FED-STD-191. The breaking strength of the finished coated fabric shall be a minimum of 53 N/mm in warp and fill.

4.6.2 Tearing strength. The finished coated fabric shall be tested in accordance with ASTM D2261. The tearing strength of the finished coated fabric shall be a minimum of 72 Newtons in warp and fill.

4.6.3 Coating adhesion. Life rafts fabricated from thermoplastic materials and using RF welding technology for seams and attachments shall undergo this test. The coating adhesion shall be determined by using a suitable air-cure adhesive to bond one face of the coated fabric to the other face. The minimum adhesion shall be 2.8 N/mm with the coating compound separating off the base fabric. Testing parameters shall be in accordance with Method 5970 of FED-STD-191.

4.6.4 Weld strength of thermoplastic coated fabrics. Life raft fabricated from thermoplastic materials and using RF welding technology for seams and attachments shall undergo this test as follows:

a. The test samples shall be in accordance with Method 5970 of FED-STD-191.

b. There shall be two samples of fabric (300 by 200 millimeters), cut with a short side parallel to the warp direction, and superimposed face to back for double coated fabrics, or coated face to coated face for single-coated fabrics.

c. The fabric samples shall be HF welded with a tool to a 10±1-millimeter width and of convenient length.

d. Twenty-five millimeter wide test specimens shall be cut transversely to the line of weld.

e. The maximum peel load shall be recorded. The minimum peel load shall be 7 N/mm for the initial break.

4.6.5 Resistance to blocking (heat). The coated fabric shall separate in 5 seconds or less when tested in accordance with the following procedure:

a. Place two 150- by 25.4-millimeter coated fabric specimens in an oven on a smooth surface in such a manner that the ends are overlapped 25.4 millimeters.

b. Place a 2.722-kilogram weight directly on the overlapped area.

c. After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

d. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down.

e. Suspend a 113.4-gram weight from the free end of the specimen.

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f. Record time required to separate.

4.6.6 Porosity test. The coated fabric shall not have any bubbles in 5 minutes when tested as follows:

a. Porosity shall be determined by a test jig as shown on figure 2.

b. The test specimen shall have a diameter of not less than 325 millimeters.

c. The specimen shall be placed, outer side up, on the holder and the plate collar tightly bolted thereon. Care shall be taken to ensure that there is no leakage at the seal.

d. A sufficient quantity of water shall be poured on the specimen to keep it completely covered under maximum pressure.

e. Pressure shall be increased to 70 kPag.

f. Air bubbles on the surface of the specimen produced by air pressure closing the spaces between the fabric and the coating shall be removed.

g. The pressure shall be held at 70 kPag for 5.0±0.1 minute.

h. At the end of the 5-minute period, the specimen shall be examined for evidence of bubble formation, indicating leakage.

i. Unless otherwise specified (see 6.2), 5 specimens from each inspection lot shall be tested.

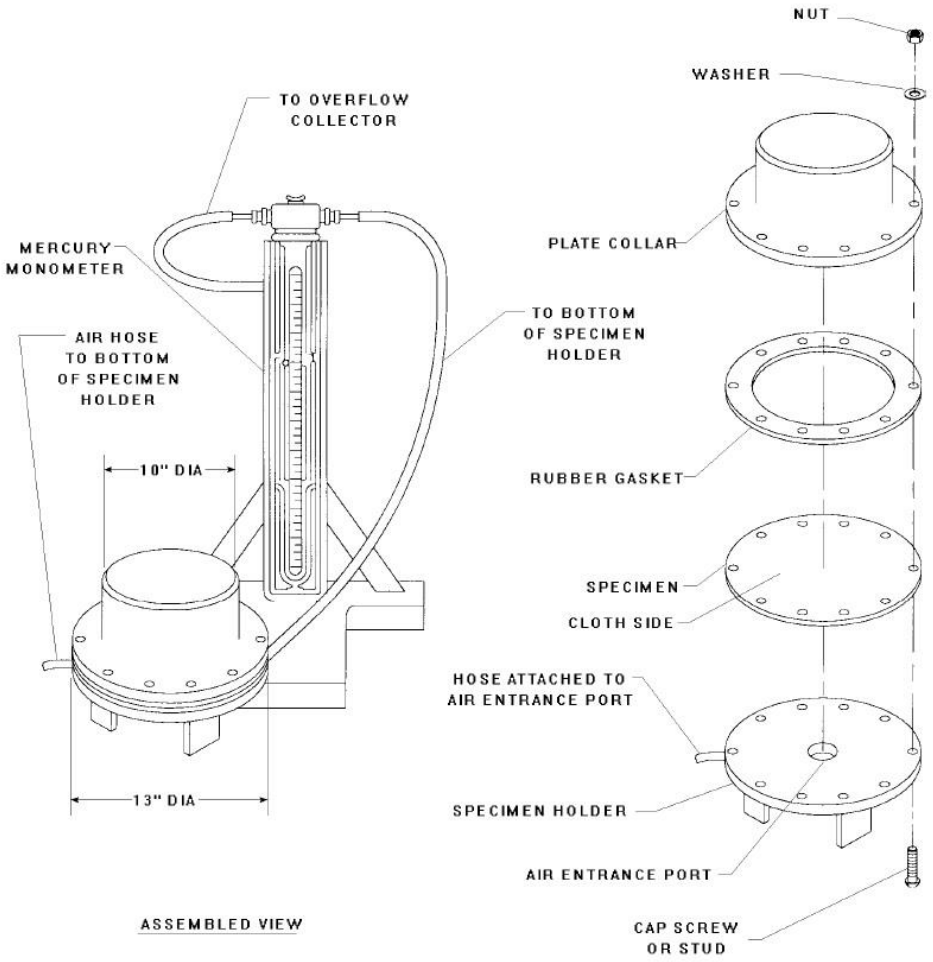


FIGURE 2. Air leakage inspection apparatus.

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4.6.7 Coating adhesion after exposure to water.

a. Three test samples shall be made in accordance with Test Method 5970 of FED-STD-191, except the adhesive system used for the samples shall be the same system used for production parts.

b. Test specimens for coating adhesion shall be immersed in 3 percent sodium chloride water solution at 60±1 °C for 96±1 hour.

c. The edges of the test specimens shall not be sealed.

d. The test specimens shall be removed from the warm water and immediately placed in water at room temperature.

e. Each determination shall be completed within 3 minutes of the time of removal from the water. Minimum adhesion results shall be 1.75 N/mm of width.

4.6.8 Puncture. The coated fabric shall have a puncture resistance of 360 Newtons when tested as follows:

a. The test shall be conducted in accordance with Method 5120 of FED-STD-191, except that the ring clamp mechanism shall have an internal diameter of 76 millimeters, and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of 7.93±0.250 millimeter and a thickness of 0.79±0.10 millimeter at the extreme tip.

b. The piercing tip edges shall be rounded to a 0.25-millimeter radius.

c. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees.

d. The average of three test specimens shall be reported.

4.6.9 Fungus. The coated fabric shall be tested in accordance with Method 508-4 of MIL-STD-810, except that the test specimen shall not be sterilized before being subjected to the fungus. The coated fabric shall maintain a minimum of 80 percent of original breaking strength and a coating adhesion of 1.75 N/mm.

4.6.10 Resistance to oil. The coated fabric shall be tested in accordance with the following method:

a. The test shall be carried out not less than 16 hours after curing.

b. Each coated face shall be tested during 22 hours at 20±1 °C.

c. The test specimen shall be folded over so that the surfaces are pressed together and there are no signs of residual tackiness.

d. With the test specimen opened out, a single pass of the finger over the exposed surface shall not produce smearing.

e. Disk specimen for testing shall be at least 70 millimeters in diameter. A typical apparatus required is shown in figure 3. It consists of a base-plate (A) and an open-ended cylindrical chamber (B), which is held tightly against the test specimen (C) by wing-nut (D). The wing-nuts are mounted on bolts (E). A hole with a diameter of approximately 30 millimeters shall be made in the base-plate for the examination of the surface not in contact with the liquid. During the test, the opening in the top of the chamber is closed by a close fitting plug (P).

f. Test specimens in the “As Received” condition shall be conditioned for not less than 3 hours at 20±2 °C immediately before testing.

g. The test specimens shall be placed in the apparatus as indicated on figure 3. The chamber of the apparatus is then filled with the test liquid to a depth of approximately 20 millimeters and the plug (P) inserted. The required temperature and the duration shall be 20±2 °C for 22 hours.

h. At the end of the contact period the test liquid is removed and the test specimen is released. Any surplus liquid is removed from the surface of the test specimen by blotting with filter paper or a textile fabric which does not deposit lint.

i. The test specimen shall be folded over so that the surfaces are pressed together to see if there are any signs of residual tackiness.

j. With the test specimen opened out, a single pass of the finger over the exposed surface shall not produce smearing.

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k. The liquid to be used as a standard test oil shall be IRM 901 Oil in accordance with ASTM D471.

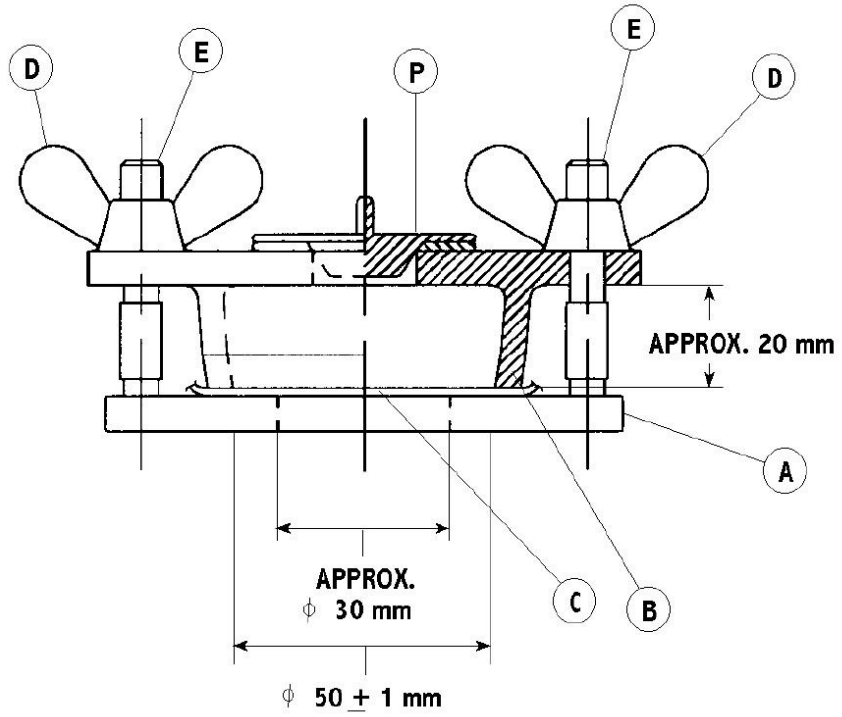


FIGURE 3. Apparatus for test with liquid on one surface only.

4.6.11 Hydrolysis resistance. When tested by the following test methods, performance values shall be achieved:

a. Coating adhesion – 70 percent of original value.

b. Blocking resistance – separation in 5 seconds when tested as follows:

(1) Place two coated fabric specimens 150 by 25.4 millimeters in an oven on a smooth surface in such a manner that the ends are overlapped 25.4 millimeters.

(2) Place a 2.722-kilogram weight directly on the overlapped area.

(3) After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

(4) Attach one end of the specimen in a suitable clamping device allowing the free end to hang down.

(5) Suspend a 113.4-gram weight from the free end of the specimen. Record the time required to separate.

c. Weld strength – 70 percent of recorded values when tested as follows:

(1) Two samples of fabric (300 by 200 millimeters), cut with a short side parallel to the warp direction, shall be superimposed face to back for double-coated fabrics or coated face to coated face for single-coated fabrics.

(2) The fabric samples shall be HF welded with a tool to a 10±1-millimeter width and of convenient length.

(3) Twenty-five millimeter wide test specimens shall be cut transversely to the line of weld.

(4) The test samples shall be in accordance with Method 5970 of FED-STD-191.

(5) The maximum peel load shall be recorded.

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4.6.12 Resistance to accelerated weathering. The exterior side of the coated fabric and representative seam samples shall be exposed in an Atlas Weatherometer or equivalent in accordance with Method 5804 of FED-STD-191. Xenon Arc or Carbon Arc Weatherometer may also be used. The cycle shall be 102 minutes of light and 18 minutes of spray during a 2-hour interval, repeated 50 times over a total time of 100 hours.

a. Coating shall not crack when the material is bent, more heavily coated side out, over a 3-millimeter diameter rod, after exposure for 100 hours without "Corex D" filters. Only one side of the fabric shall be tested, which shall be the side facing the arc in the testing apparatus. After weathering, the coated fabric and representative seam samples shall show a loss in initial tensile strength not exceeding 10 percent when tested in accordance with Method 5102 of FED-STD-191.

b. The resistance to blocking (heat) shall be tested as follows:

(1) Place two coated fabric specimens 150 by 25.4 millimeters in an oven, on a smooth surface, in such a manner that the ends are overlapped 25.4 millimeters.

(2) Place a 2.722-kilogram weight directly on the overlapped area.

(3) After conditioning at a temperature of 70±2 °C for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 22.7±3 °C.

(4) Attach one end of the specimen in a suitable clamping device, allowing the free end to hang down.

(5) Suspend a 113.4-gram weight from the free end of the specimen.

(6) Record the time required to separate.

c. Single-coated fabrics shall be subjected to a coating adhesion test (see 4.6.3), after weatherometer, and shall have a minimum of 70 percent of the coating adhesion.

d. Upon completion of steps a and b above, the test sample shall be folded and shall exhibit no cracks, delamination, or visual deterioration when samples are inspected under a magnification of 2 times.

4.6.13 Abrasion resistance. The exterior side of the inflatable fabric shall withstand a minimum of 25,000 cycles of chafing without exposure of the fabric substrate when the fabric is coated on one side only. When the fabric is coated on both sides and the inside coating is at least equal to or greater than the exterior coating, the exterior side shall withstand a minimum of 15,000 cycles without exposure of fabric substrate. Method 5306 of FED-STD-191 using a CS-17 wheel with 500-gram weights shall be used in testing the material.

4.6.14 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, and closures, chafing or reinforcing patches, and coated fabric flanges of valves) shall be considered as seams. They shall be subjected to all seam tests specified in table I (except for chafing patches, which will only be tested for peel) as follows:

a. The average breaking strength of five specimens for each type of seam for each test shall be compared to table I for conformance.

b. Breaking strength specimens shall be 50 millimeters wide (parallel to the seam) and shall extend (perpendicular to the seam) 75 millimeters beyond both edges of the seam.

c. For tests conducted after immersion in a fluid, no part of the test specimens shall be coated or covered during the water immersion periods.

d. Specimens shall be cooled in the immersion fluids at 23±3 °C for 30 to 90 minutes before testing.

e. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids.

f. The average peel adhesion strength of two specimens for each type seam shall be compared to table I for conformance.

g. Peel adhesion specimens shall be of sufficient length to determine both the initial and after water adhesion values on the same specimen.

4.6.15 Dead load shear resistance. The dead load shear resistance of the seams shall be tested in accordance with the following procedure:

a. Three specimens shall be tested for each seam type.

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b. The test specimens shall be 25.4±0.050 millimeters wide (parallel to the seam) and coated fabric shall extend a minimum of 75 millimeters (perpendicular to the seam) on each side of the seam.

c. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage.

d. Each specimen shall be subjected to a constant (dead load) force of 23±0.25 kilograms, at 93±3 °C.

e. After 24 hours, each specimen shall be examined while still under load for signs of slippage or separation as stated in table V.

TABLE V. Characteristics of seams.

Property	Requirements	Test Method
Breaking Strength (initial)	53 N/mm (min)	ASTM D751[1/] and as specified herein
Breaking Strength – after immersion in distilled water at 71±1 °C for the following duration: - 14 days - 42 days	 49 N/mm 25 N/mm	ASTM D751, ASTM D471, and as specified herein
Dead Load Shear Resistance – under 50 lb/in in stress at 200 °F for 24 hours	2.5 mm slippage (max)	As specified herein
Peel Adhesion (initial)	3.5 N/mm (min)	ASTM D413 machine method and as specified
Peel Adhesion – after immersion in distilled water at 71±1 °C for the following duration. - 14 days - 42 days	 1.75N/mm 1.40 N/mm	ASTM D413 machine method, ASTM D471, and as specified herein
NOTE: [1]/ Except that the specimen shall be 50.8 millimeters in width.		

4.7 Life raft tests.

4.7.1 Air retention test. Each life raft shall pass an air retention test under the following conditions:

a. The testing period shall be 4 hours, with a maximum allowable pressure drop of 2.5 percent after compensation for ambient temperature and barometric pressure changes.

b. For each degree Celsius rise in temperature, 0.385 kilopascal shall be subtracted from the final pressure reading. For each degree Celsius drop in temperature, 0.385 kilopascal shall be added to the final pressure reading.

c. For each millimeter of mercury rise in barometric pressure during the 4-hour period, 0.133 kilopascal shall be added to the final temperature-corrected pressure reading. For each millimeter of mercury drop in barometric pressure, 0.133 kilopascal shall be subtracted from the final temperature-corrected pressure reading.

d. One inflatable life raft from each lot of less than 30 life rafts and 2 from each lot of 30 or more life rafts shall pass the requirements of 4.7.1. If any life raft fails this test:

(1) The reason for the failure shall be determined.

(2) Each life raft in the lot shall be examined for the defect.

(3) The lot test shall be repeated, including random selection of the life raft or life rafts to be tested.

4.7.2 Jump test. Thirteen (Mark-7) or 25 (Mark-8) of the jumps shall be with the canopy erect and 12 (Mark-7) or 25 (Mark-8) of the jumps with the canopy furled or deflated. The test subject shall not weigh less than 75 kilograms. If a “suitable and equivalent mass” is used, it shall be equipped with the shoes described in Part 1, Regulation 5.2.1 of IMO Resolution MSC.81(70) arranged to strike the life raft first.

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4.7.3 Inflation actuation test – ambient temperature. The Mark-7 and Mark-8 life rafts shall be tested to validate the ability of the life rafts to inflate by sea painter actuation of the inflation system. This test shall be performed at a temperature range of 18 to 30 °C and use the following steps to execute.

a. The rubber plug shall be removed from the sea painter bung assembly.

b. The sea painter shall be pulled from the sea painter bung assembly until resistance is felt.

c. A bight shall be tied on to the sea painter a safe distance from the life raft so as test personnel are not struck by the inflating life raft.

d. A calibrated full force gauge set to peak hold shall be attached to the bight in the sea painter.

e. Time shall be started on a stop watch at which time the test person shall exert pull on the pull force gauge attached to the bight in the sea painter. The pull shall be steady and kept parallel with the floor and centerline of the life raft container until the inflation system is actuated.

f. The inflation of the life raft shall be observed to ensure the life raft fully inflates with: arch and canopy up, all boarding ramps inflated, pressure relief valves venting, exterior canopy lamp(s) flashing, and interior canopy lamp(s) illuminated. When the life raft is full inflated, timing with stop watch shall halt.

g. After the pressure relieve valves have stopped venting and the temperature of the life raft stabilized, measure the pressure in the hull chambers.

h. A successful inflation test results in complete inflation in less than 30 seconds (see 3.11.1) with an actuation pull force of 150 Newtons or less (see 3.6.1.b), hull chambers air pressure attaining the life raft working pressure (see 3.6.4), and exterior canopy lamp(s) (see 3.5.3) and interior canopy lamps(s) (see 3.5.4) illuminated.

4.7.4 Inflation actuation test – cold temperature extreme. The Mark-7 and Mark-8 life rafts shall be tested to validate the ability of the life rafts to inflate by sea painter actuation of the inflation system. This test shall be performed at a temperature of -30 °C using the following steps:

a. The life raft shall be cold soaked to a temperature of -30°C for a period not less than 24 hours.

b. The life raft shall be removed from the environmental chamber.

c. The rubber plug shall be removed from the sea painter bung assembly.

d. The sea painter shall be pulled from the sea painter bung assembly until resistance is felt.

e. A bight shall be tied on to the sea painter a safe distance from the life raft so as test personnel are not struck by the inflating life raft.

f. A calibrated full force gauge set to peak hold shall be attached to the bight in the sea painter.

g. Time shall be started on a stop watch at which time the test person shall exert pull on the pull force gauge attached to the bight in the sea painter. The pull shall be steady and kept parallel with the floor and centerline of the life raft container until the inflation system is actuated.

h. The inflation of the life raft shall be observed to ensure the life raft fully inflates with:

(1) Arch and canopy up.

(2) All boarding ramps inflated.

(3) Pressure relief valves venting.

(4) Exterior canopy lamp(s) flashing.

(5) Interior canopy lamp(s) illuminated.

When the life raft is full inflated, timing with stop watch shall halt.

i. After the pressure relieve valves have stopped venting and the temperature of the life raft stabilized, measure the pressure in the hull chambers.

j. A successful inflation test results in complete inflation in less than 30 seconds (see 3.11.1) with an actuation pull force of 150 Newtons or less (see 3.6.1.b), hull chamber air pressure attaining the life raft working pressure (see 3.6.4), and exterior canopy lamp(s) (see 3.5.3) and interior canopy lamps(s) (see 3.5.4) illuminated.

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4.7.5 Over-pressure test. The hull and arch chambers shall be subjected to a 300 percent working pressure test (see 3.11.2) conducted as follows:

a. All pressure relief valves shall be disabled with a poppet retainer (see 3.6.3). The pressure of each hull and arch chamber shall be increased to 300 percent the working pressure of the life raft and held for 5 minutes. During that time, no failure of the air retention integrity of the chambers shall occur.

b. The poppet retainers shall then be removed to allow the chamber to vent the additional pressure.

c. No failure of the seams, fabric, or valves shall occur during the test.

4.7.6 Mooring-out test. The life raft shall be loaded with a mass equal to the total number of personnel the life raft is rated (Mark-7: 1,875 kilograms, Mark-8: 3,750 kilograms) and its equipment. The life raft shall be moored at sea or in a seawater harbor. The life raft may be topped-off once per day with the manual inflation pump; however, it should retain its shape over any 24-hour period. Initial inflation shall be compressed air. The life raft shall remain afloat for 30 days without any damage or degradation that would impair its performance.

4.7.7 Loading and seating test. The life raft shall be tested for loading and seating in accordance with Part 1, Regulation 5.7 of IMO Resolution MSC.81(70). The persons used to determine seating capacity shall wear insulated buoyant immersion suits rather than lifejackets.

4.7.8 Canopy closure test. Canopy closure shall be tested in accordance with Part 1, Regulation 5.12 of IMO Resolution MSC.81(70). The water accumulated within the life raft at the end of the testing shall not exceed 20 liters.

4.7.9 Lift-out force test. The life raft shall be subjected to a lift-out force test as follows:

a. The test shall be conducted in a pool or body of water where the wind is less than 8 knots and the current is less than 1 knot. The life raft shall be tested in its ‘light condition” fully equipped with no personnel.

b. The life raft shall be inflated and placed in the water at the test site. Each pressure relief valve shall be made inoperative. The stability appendages shall be allowed to deploy. If the stability appendages do not deploy on their own, the life raft may be agitated until the appendages deploy. The stability appendages shall not be manually pulled open or arranged for this test.

c. The life raft shall be subjected to an upward vertical force applied at the outer edge of one of the main buoyancy chambers by means of a lifting bridle. The lifting bridle may incorporate existing towing, lifeline, or other attachments, or may be specially constructed for this test.

d. If the life raft has a waterplane that is a circle or other symmetrical shape, the lifting bridle shall be located at the point where the resisting moment created by the inflation cylinders and equipment packs is minimized.

e. If the life raft has a waterplane that is other than a circle or symmetrical shape, it shall be lifted at each of two locations; once at one end of the major axis of the waterplane and once at one end of the minor axis of the waterplane. The end of the axis selected for attachment of the lifting bridle shall be the end where the resisting moment created by the inflation cylinders and equipment packs is minimized.

f. The vertical force shall be applied so that the lifting bridle rises at a speed of 1.67B meters/minute (m/min) (+5 percent), where “B” is the length in meters of the axis on which the lifting bridle is placed. The lifting bridle shall be lifted a distance of at least (B) (sin 20 degrees). The force applied to the lifting bridle shall be continuously measured as the edge of the life raft is lifted. Once the lifting bridle has been raised, the required distance the life raft shall be lowered back to the surface of the water.

g. The peak force measured during the lifting of the life raft shall be at least 3315 Newtons. The procedure shall be repeated up to a total of 3 runs at the required lifting rate if the minimum lift-out force is not observed on a particular trial. Test results at lifting rates which are not within the limits of the required lifting rate shall be disregarded.

h. After the life raft has been lowered back to the surface of the water, it shall assume its design shape and shall show no evidence of damage or leakage.

4.7.10 Towing test. The life raft, with full complement of persons and equipment, shall be able to be towed at a speed of 3 knots for 1 kilometer when subjected to a towing test in accordance with Part 1, Regulation 5.4 of IMO Resolution MSC.81(70).

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4.7.11 Drop test. The life raft, container, and survival equipment shall survive a drop of 23 meters from the water surface with no damage to the container, life raft, and survival equipment and remain afloat for 30 minutes when subjected to a drop test in accordance with Part 1, Regulation 5.1 of IMO Resolution MSC.81(70) except the drop height shall be 23 meters from water surface.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The life rafts described herein are intended for shipboard installation to be used in the event the ship shall be abandoned at sea. These life rafts are military unique since they have a container, sea painter, flares, survival manual, inflation system, and a government-furnished manual reverse osmosis desalinator. Additionally, they have a military specific drop height and certification periodicity.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced when required (see 2.1, 2.2, and 2.3).
c. When first article inspection is required (see 3.1).
d. Requirements for repair or rework (see 3.4.14).
e. Government-furnished property (see 3.7.9, 3.7.17, 3.7.19, and 6.3).
f. Training material and maintenance instruction requirements (see 3.12).
g. Service Manual requirements (see 3.13.2).
h. When other than 5 specimens are tested as part of the porosity test (see 4.6.6).
i. Packaging requirements and additional packaging considerations (see 5.1 and 6.8).
j. Responsibility for inspection and compliance (see 6.6).
k. Requirements for contractor certification (see 6.7).
l. Requirements for documentation (see 6.8).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.7.9, 3.7.17, and 3.7.19 (see 6.2).

6.4 Metric products. Products manufactured to inch/pound dimensions will be considered on an equal basis with those manufactured using metric units, provided they fall within the tolerances specified using conversion tables contained in IEEE/ASTM SI10 and all other requirements of this specification are met.

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6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for Type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.6 Responsibility for inspection and compliance. Requirements for inspection and compliance are as specified in the contract (see 6.2).

6.7 Contractor certification. The requirements for contractor certification are as specified (see 6.2). The Contractor should certify and maintain substantiating evidence that the product offered meets the characteristics of Section 3.0 of the specification and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as will be otherwise provided for under the provisions of the contract.

6.8 Documentation. The requirements for documentation are as specified (see 6.2). Each life raft should be provided with two copies of documentation, which include: life raft serial number, manufacture and certification dates, and serial numbers of: inflation cylinders, inflation valves, the manual reverse osmosis desalinator (MROD), and the lot number of the GFM signal kit. The Contractor should retain on file, for a period of 5 years, records which contain material test results (production lot test) and test records. The Contractor should provide periodic reports, quarterly at a minimum, with life raft certification data on new life rafts to NSWCCD Detachment Norfolk.

6.9 Additional packaging considerations for contract application. The following additional packaging considerations will be addressed in the contract, as specified (see 6.2).

6.9.1 Packing. Each life raft should be packed in an enclosed wood or cardboard box crate acceptable to the common carrier, which will ensure safe delivery to destination in a satisfactory condition. Containers, packing, and method of shipment should comply with Uniform Freight Classification or National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation. Method of packing should facilitate handling and storage by fork lift trucks and should be capable of multiple handling and storage under favorable conditions for a minimum of 1 year. Crate or box should have tamper seal(s) on them to indicate whether or not the packaging has been opened.

6.9.2 Marking. Interior packages and exterior shipping containers, including bar coding, should be marked to identify items being delivered. The life raft, and any other included item with shelf-life requirements, should include the date of manufacturer (month and year). The shipping container should be marked with the month and year of shipment and the life raft serial number.

6.9.3 Special marking. Filled cylinders, interior containers housing same, and the exterior shipping container should contain a green label in accordance with 49 CFR 170 Subpart 179-199. In addition, the above should be marked "NON-FLAMMABLE COMPRESSED GAS" or "COMPRESSED AIR". The size and color of the markings should conform to DOT Rules and Regulations as specified herein. The markings should be conspicuous and legible.

6.9.4 Hazardous material marking. Interior packages and items, including each life raft container and exterior shipping container, should be marked and labelled to identify the compressed air or gas signal device kit, or any other hazardous material contained with the packed life raft. Markings and labels should conform to 49 CFR 171-178.

6.9.5 Shipping crate markings. Minimum markings should include the name and address of consignee and consignor and any applicable and any precautionary markings. Contract data markings should be placed below the identification markings.

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6.9.6 Methods of marking. Marking should be accomplished by use of labels, stamping, stenciling, machine printing, or tagging. Lithographing, silk screening, photo marking, embossing, decals, transfers and bar coding of laser marking may also be used. Hand lettering or hand printing should not be used.

a. Size of lettering. Lettering for all markings should be capital letters of equal height as specified herein, so as to be clearly legible and proportional to the available marking space of the container. Unless otherwise specified herein, stenciled lettering for identification, contract data, and special markings should be not less than 6 millimeters nor more than 25 millimeters height. Address, identification, and contract data markings on plastic, fabric, paper tags, or paper labels should not be less than 2.5 millimeters.

b. Labels. Labels should be of a water-resistant grade of paper, film, fabric, or plastic, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material should be such as to permit flexibility. Labels should have a finish capable of withstanding normal handling during shipment and storage. Labels should have a finish suitable for printing and writing on with ink without feathering or spreading. The adhesive should be a pressure-sensitive permanent type. It should be homogeneous and should be coated in a smooth layer on the unprinted side of the label. The adhesive should require no solvent, heat, or other preparation prior to application. Adhesive should be of a type that will adhere to metal, plastic, or fiberboard surfaces under high or low temperatures. The applied label should remain securely in position under anticipated conditions of handling, shipment, and storage. Labels should be fade resistant.

6.10 First article inspection. When a first article inspection is required (see 6.2), the contracting officer will include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and report, and disposition of first articles. Invitations for bids will provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, will furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.11 Subject term (key word) listing.

Abandon ship

Lifesaving

Survival

Preparing activity:
Navy – SH
(Project 4220-2015-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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